

EXP-3-Record-Histogram processing

name : jana shravin S

reg no: 212224243003

In [14]:

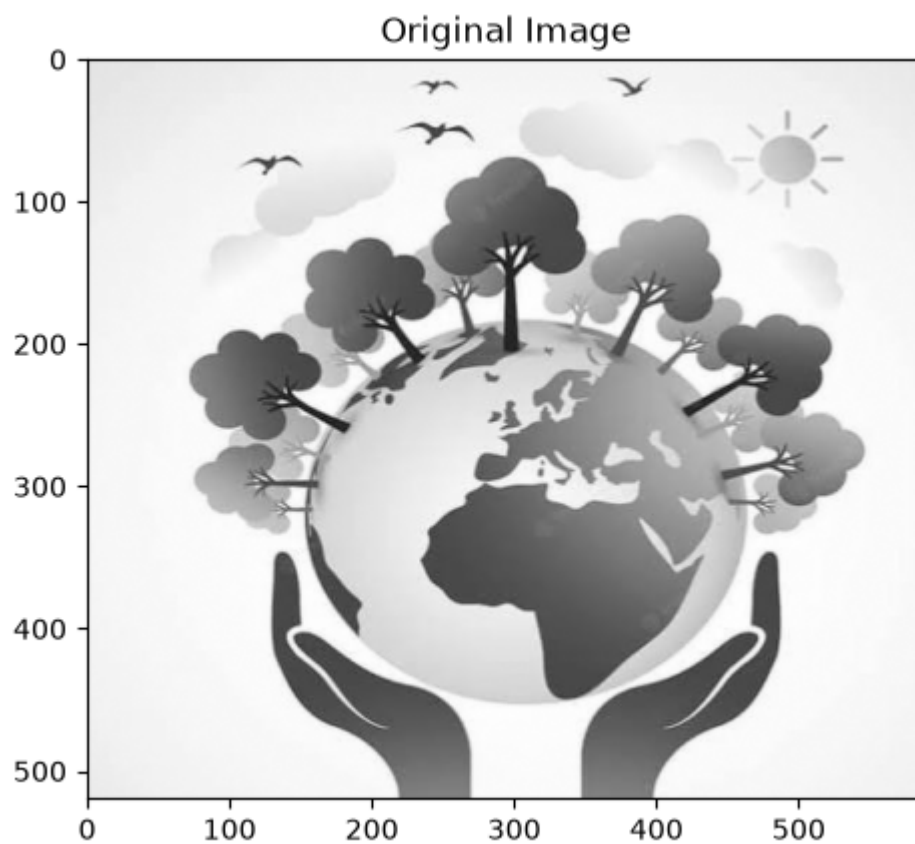
```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

In [15]:

```
img = cv2.imread('saveetha.jpg', cv2.IMREAD_GRAYSCALE)
```

In [16]:

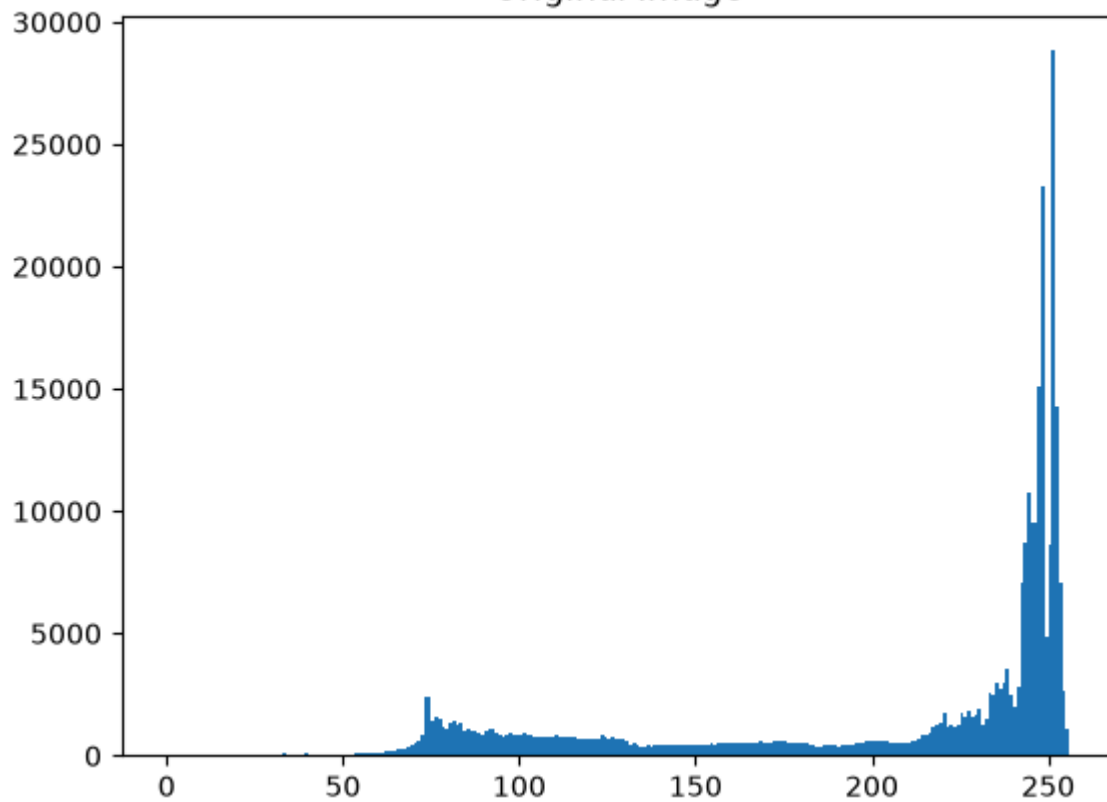
```
plt.imshow(img, cmap='gray')
plt.title('Original Image')
plt.show()
```



In [17]:

```
plt.hist(img.ravel(),256,range = [0, 256]);
plt.title('Original Image')
plt.show()
```

Original Image



In [18]:

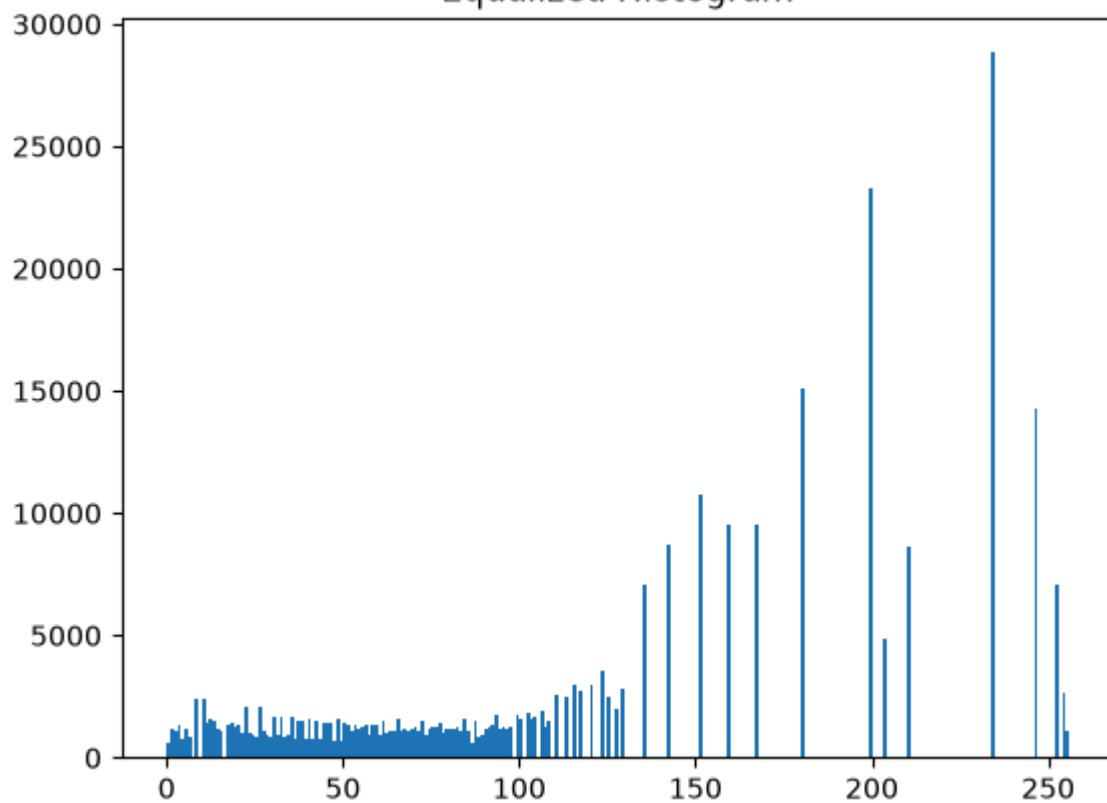
```
img_eq = cv2.equalizeHist(img)
```

In [19]:

```
plt.hist(img_eq.ravel(), 256, range = [0, 256])  
plt.title('Equalized Histogram')
```

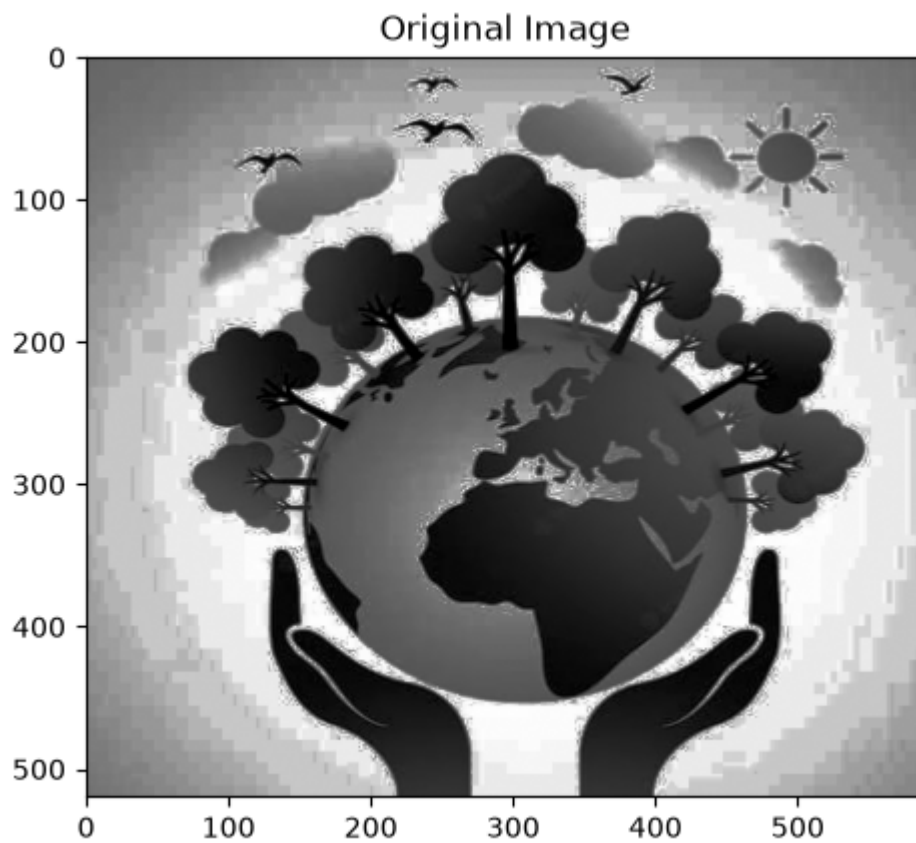
Text(0.5, 1.0, 'Equalized Histogram')

Equalized Histogram



In [20]:

```
plt.imshow(img_eq, cmap='gray')  
plt.title('Original Image')  
plt.show()
```



In [21]:

```
img = cv2.imread('saveetha.jpg', cv2.IMREAD_COLOR)
```

In [22]:

```
img_hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
```

In [23]:

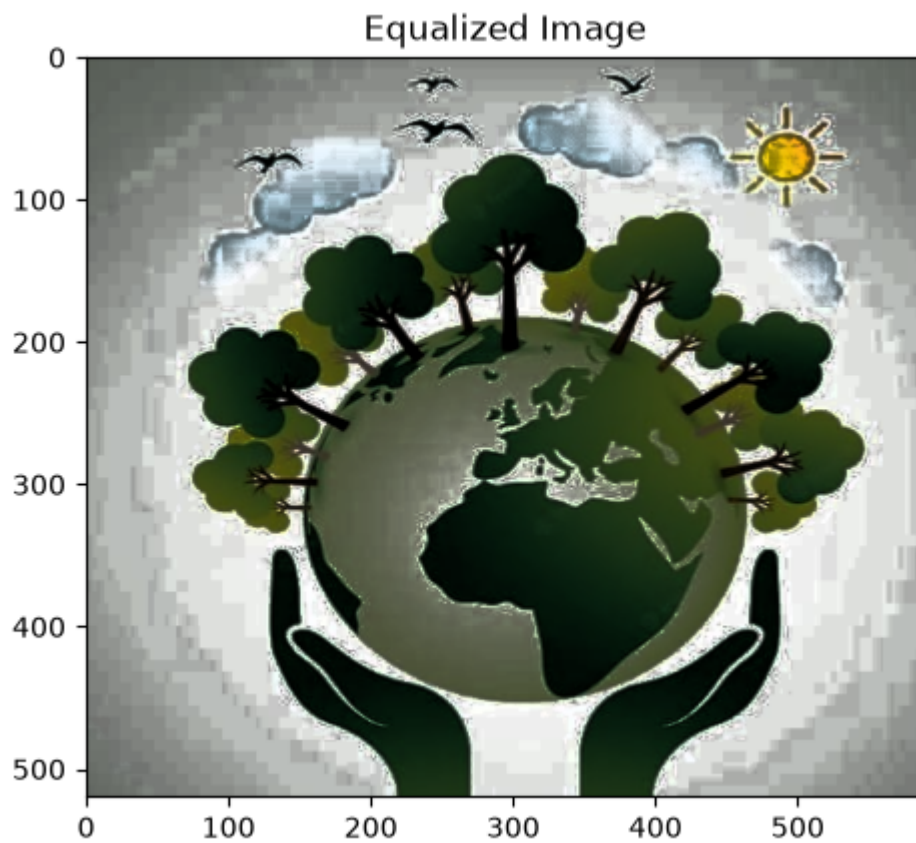
```
img_hsv[:, :, 2] = cv2.equalizeHist(img_hsv[:, :, 2])
```

In [24]:

```
img_eq = cv2.cvtColor(img_hsv, cv2.COLOR_HSV2BGR)
```

In [25]:

```
plt.imshow(img_eq[:, :, ::-1]); plt.title('Equalized Image'); plt.show()
```

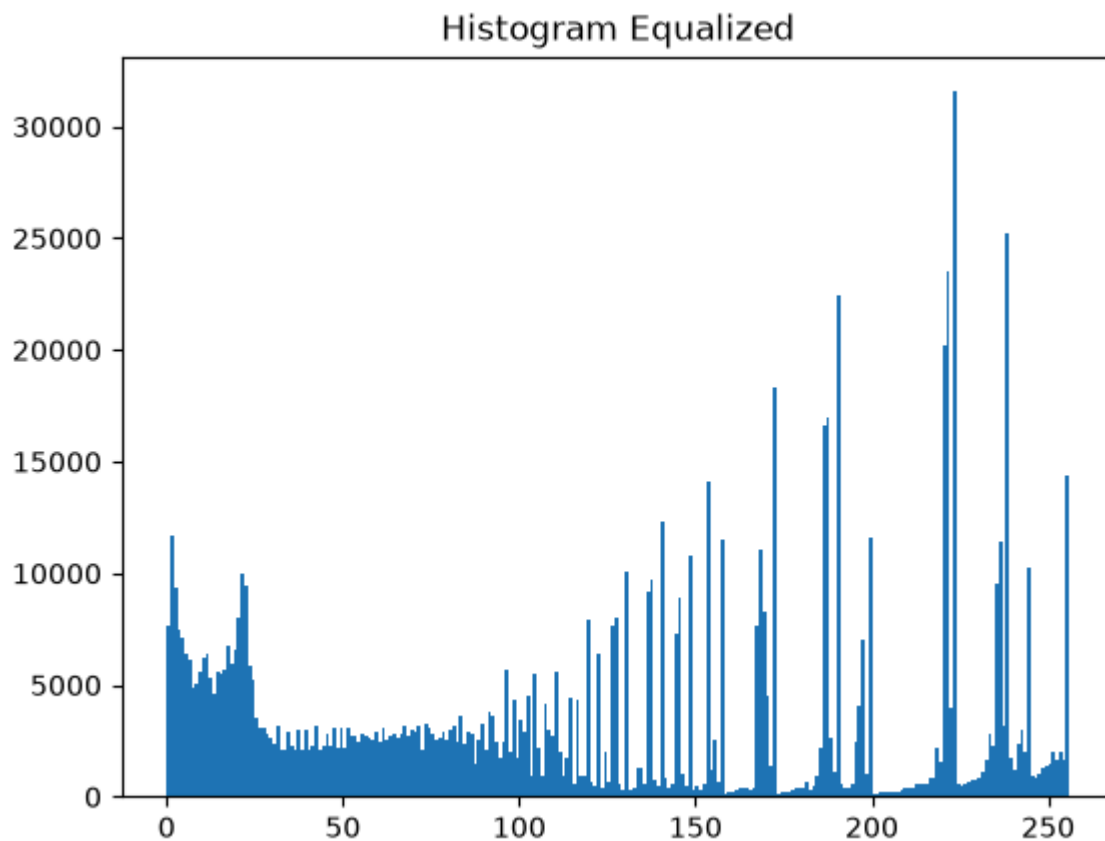


In [26]:

```
plt.hist(img_eq.ravel(),256,range = [0, 256]); plt.title('Histogram Equalized');plt.show()
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[26], line 1
----> 1 plt.hist(img_eq.ravel(),256,range = [0, 256]); plt.title('Histogram Equalized');plt.show()

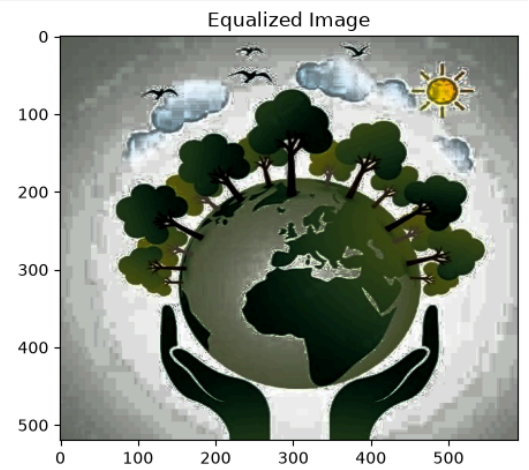
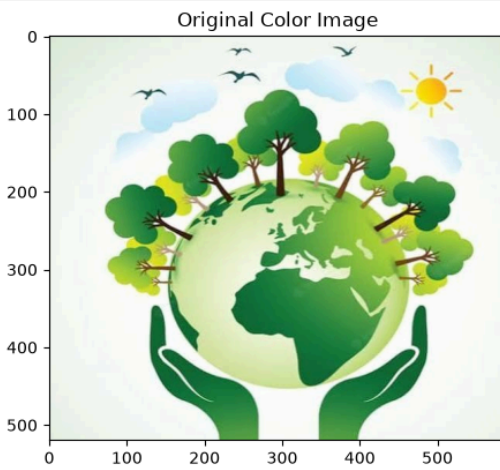
AttributeError: module 'matplotlib.pyplot' has no attribute 'show'
```



In [31]:

```
plt.figure(figsize = (20,10))
plt.subplot(221); plt.imshow(img[:, :, ::-1]); plt.title('Original Color Image')
```

```
plt.subplot(222); plt.imshow(img_eq[:, :, ::-1]); plt.title('Equalized Image')
plt.show()
```



In [32]:

```
plt.figure(figsize = [15,4])
plt.subplot(121); plt.hist(img.ravel(),256,range = [0, 256]); plt.title('Original Image')
plt.subplot(122); plt.hist(img_eq.ravel(),256,range = [0, 256]); plt.title('Histogram Equalized')
```

Text(0.5, 1.0, 'Histogram Equalized')

